

Gorilla Engine

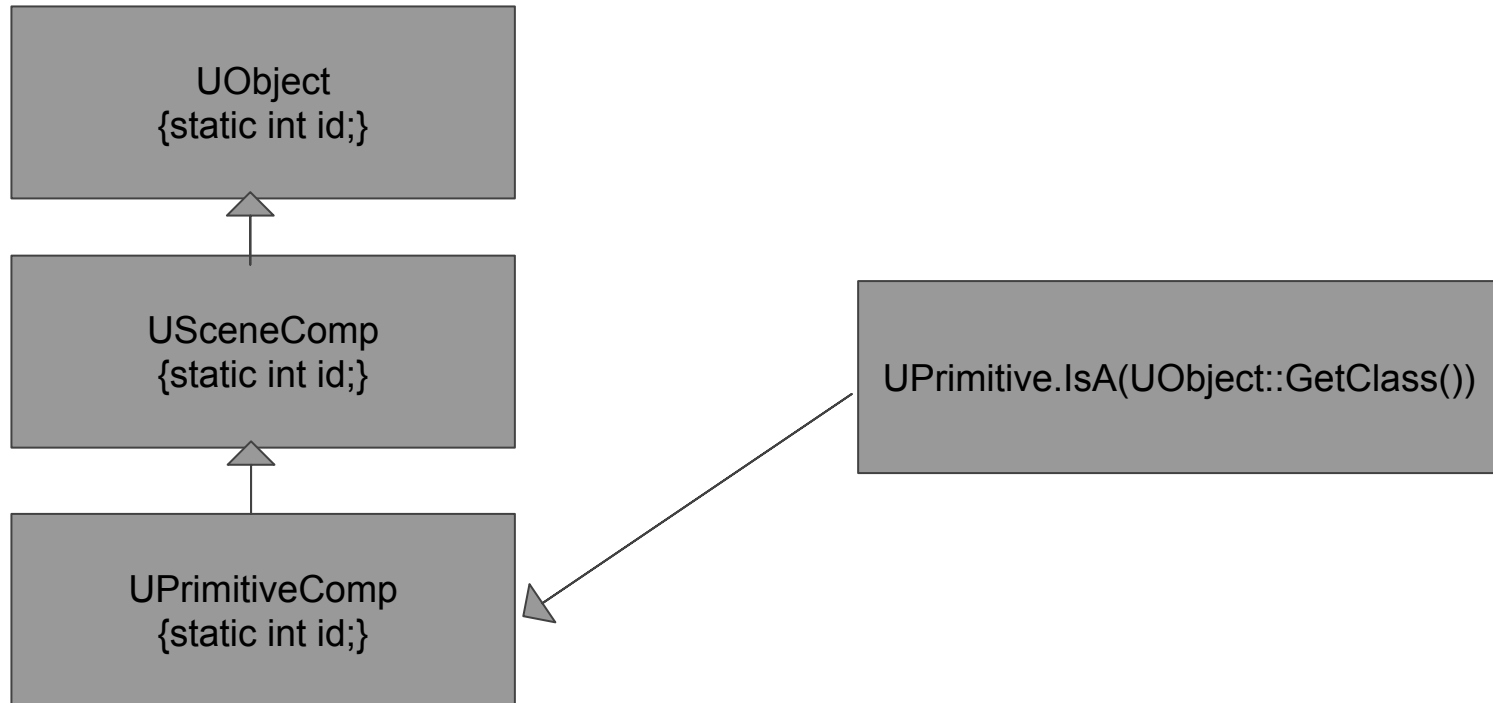
4팀 김태현 김형도 이호진 국동진

목차

- 시연
- RTTI와 Factory
- 전체 구조
 - World Update
 - FLevel

Demonstration

RTTI



Update Loop

UWorld

UScene* PermanentScene
UScene* ActiveScene

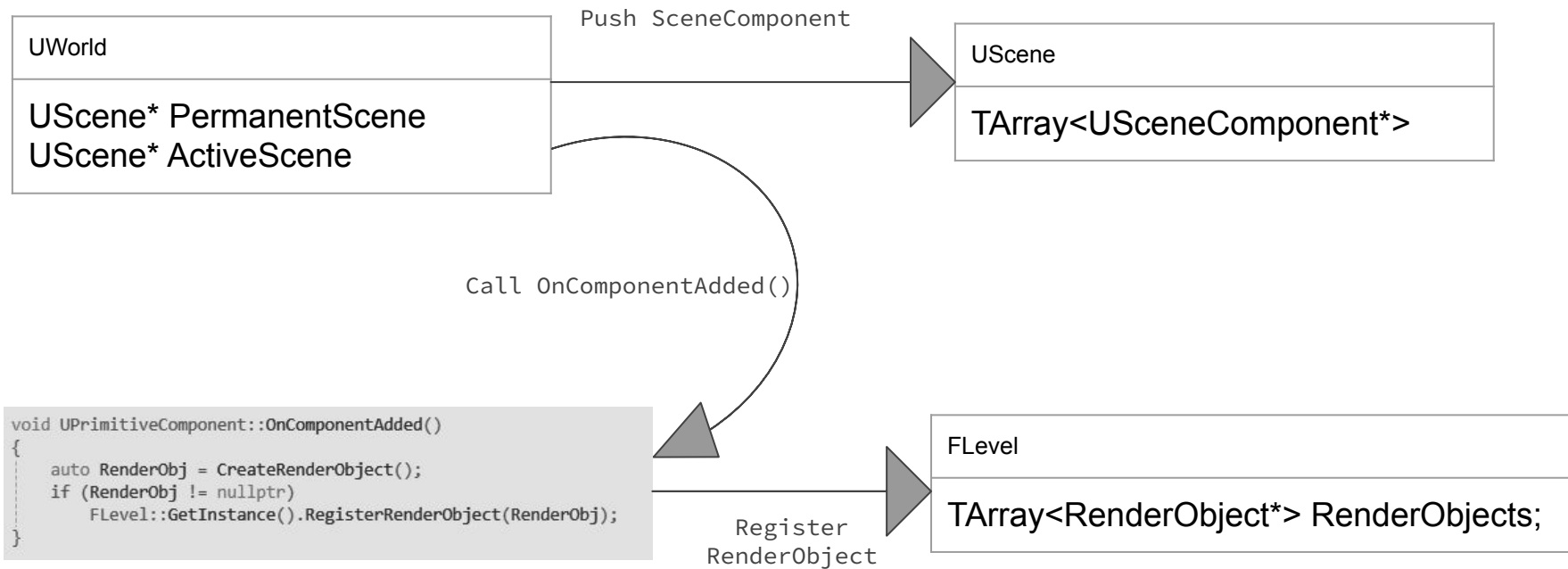


```
void UWorld::Update(float deltaTime)
{
    permanentScene->Update(deltaTime);
    activeScene->Update(deltaTime);
}
```



```
void UScene::Update(float DeltaTime)
{
    for (USceneComponent* sc : SceneComponents)
    {
        sc->Update(DeltaTime);
    }
}
```

Rendering Loop



Rendering Loop

```
void UWorld::OnBeforeRender() const
{
    auto sceneComponents = activeScene->SceneComponents;
    for (USceneComponent* s : sceneComponents)
    {
        UPrimitiveComponent* primitive = Cast<UPrimitiveComponent>(s);
        if (primitive != nullptr)
        {
            if (primitive->GetRenderObject()->bIsDirty)
                primitive->ResolveRenderStateDirty();
        }
    }
}
```

```
void UPrimitiveComponent::UpdateRenderObject()
{
    ResourceManager* RM = ResourceManager::GetInstance();
    RenderObj->World = GetRelativeMatrix();
    RenderObj->bIsSelected = bIsSelected;
}

void UPrimitiveComponent::ResolveRenderStateDirty()
{
    UpdateRenderObject();
    RenderObj->bIsDirty = false;
}
```

```
auto renderObjects = &FLevel::GetInstance().RenderObjects;
for (RenderObject* RenderObj : *renderObjects) { ... }

for (RenderObject* RenderObj : NormalRenderObjects)
{
    //Camera
    ObjectConst.World = RenderObj->World;
    ObjectConst.Color = RenderObj->Color;
    ObjectConst.bIsSelected = RenderObj->bIsSelected ? 1 : 0;
    ObjectConstantBuffer.Update(DeviceContext.Get(), &ObjectConst, sizeof(FObjectConstant));

    //GPU에 ConstantBuffer가 오브젝트마다 생기는 것을 방지하기 위해 ObjectConstantBuffer를 하나만 만들어서
    RenderObj->Render(DeviceContext.Get());
}
```